

Empowering Autonomy and Agency: Exploring and Augmenting Accessible Cyber-Physical Systems

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ABSTRACT

People with Vision Impairments face various barriers in physical activities that are highly visual (e.g., cooking, visiting art museums), which strongly impact their agency, autonomy, and quality of life. Cyber-Physical Systems (CPS) provide opportunities for people with vision impairments to better interact with physical space. In this research, we first explore current practices and challenges of the adaptation of CPS by people with vision impairments in physical activities. We then establish a comprehensive design space and design guidelines for CPS to satisfy the needs of blind individuals. Finally, we integrate the user-centered design guidelines, implement CPS for people with vision impairments to reduce the barrier of interacting with physical space, and conduct deployment studies to validate our design. Overall, our research provide a roadmap of augmenting accessible CPS for people with disabilities in the physical space.

CCS CONCEPTS

• **Human-centered computing** → **Accessibility technologies; Empirical studies in accessibility; Accessibility design and evaluation methods; Ubiquitous and mobile computing systems and tools; Empirical studies in ubiquitous and mobile computing.**

KEYWORDS

Accessibility, Assistive Technology, HCI, Ubiquitous Computing, Mixed-Methods Evaluation

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1 PROBLEM STATEMENT

There are over 2.2 billion people with vision impairments in the world [1] and many physical activities, such as cooking, eating, and appreciating arts, can be difficult without visual support [20, 21]. These barriers to physical activities strongly impact the Quality of Life of people with vision impairments [15]. For example, people with vision impairments tend to have poor nutritional status due to the difficulty accessing visual information and cues during cooking processes [7, 15, 17]. The capability of engaging in these physical activities further impacts the equity, agency, and autonomy of people with vision impairments [22]. Traditional assistive technologies have provided some solutions, but they often fall short in addressing the complex needs and aspirations of blind individuals in physical activities [20–22]. Prior research uncovered that over 20% of the assistive technologies were abandoned due to many reasons, such as functionality, social interactions, and usability [26].

Cyber-Physical Systems (CPS) have emerged as a transformative technology, integrating computation, communication, and control into the physical world [10]. The intersection of CPS and specific challenges faced by blind people presents an opportunity to enhance their autonomy, agency, and overall Quality of Life. CPS can offer a range of functionalities that enable real-time sensing, data processing, and actuation [18], all crucial aspects for enhancing accessibility and independence. For example, the integration of computer vision algorithms within CPS can enable real-time object detection and recognition, facilitating obstacle avoidance and object identification for blind individuals [24]. Additionally, CPS can provide haptic feedback, audio cues, or tactile interfaces to convey information about the physical environment, allowing blind people to navigate and interact more effectively. However, limited research has explored how should CPS be designed and developed to support people with vision impairments in physical activities with further deployment.

To explore and augment accessible Cyber-Physical Systems, here are the three main research questions that my research focus on:

- RQ1: What are the current practices, challenges, and strategies involved in the adaptation of Cyber-Physical Systems by blind individuals, and how can we better understand and address their unique needs and preferences?
- RQ2: How can we establish a comprehensive design space for Cyber-Physical Systems tailored to the specific requirements of blind individuals, considering factors such as user interface design, sensory feedback modalities, and real-time object detection capabilities?
- RQ3: How can we effectively integrate user-centered design guidelines derived from user studies into the development

and implementation of Cyber-Physical Systems for blind individuals, ensuring seamless integration, improved usability, and long-term acceptance?

2 RELATED WORK

In this section, we present background and related work on 1) People with Vision Impairments and Physical Activities, and 2) Cyber-Physical Systems and Accessibility.

2.1 People with Vision Impairments and Physical Activities

People with visual impairments' Activities of Daily Living (e.g., preparing food) and Instrumental Activities of Daily Living (e.g., visiting art museums) are generally impacted by the loss of vision [6, 20, 22]. The physical barriers include the lack of awareness of space, the high density of visual information, and the complexity of the physical space [20]. For example, Jones et al. [15] conducted a survey study with 101 visually impaired people and found 65% of the participants stated that their visual impairments made cooking difficult. Due to the difficulty of cooking, Bilyk et al. [7] found that people with vision impairments tend to eat outside or prepared food, which affects healthy eating behaviors. Beyond the home environment, people with vision impairments also face various difficulties in the public physical space, such as visiting art galleries and museums [22]. The inaccessible experience impacts the quality of life, agency, and autonomy of blind people [21]. Given existing research has explored various technologies to support people with vision impairments with specific tasks (e.g., using smartphone, interacting with smartwatch), less is focused on how technology should be adopted in the physical space. Shinohara and Wobbrock [26] uncovered that the physical adoption of technology is impacted by many factors (e.g., social interactions, functionality, robustness), and over 20% of assistive technologies were abandoned by people with disabilities. Therefore, it is important to understand what are the current practices, challenges, and strategies that got involved in the adaptation of technologies by people with vision impairments in physical activities.

2.2 Cyber-Physical Systems and Accessibility

Cyber-Physical Systems were defined as a form of integration of computing systems with physical capabilities that can support humans to interact through various modalities [5]. Existing research has explored CPS for accessibility purposes, which focused on interactions with physical interfaces [11], and navigation [25]. For example, prior research also explored using computer vision [9, 11, 23, 27], voice interactions [4, 8], and 3D printed tactile marking [12, 13] to better support people with visual impairments interacting with different physical interfaces. Specifically, VizLens leveraged computer vision and crowdsourcing to enable people with visual impairments to interact with different interfaces, such as a microwave oven [11]. Guo et al. [12] further introduced a crowd-sourced fabrication pipeline to help blind people independently make physical interfaces accessible through adding a 3D-printed augmentation of tactile buttons overlaying the original panel. Given the existing research on CPS for blind individuals, many people

with vision impairments still face barriers to various physical activities (e.g., [20, 22]). It is important to systematically understand and explore the adaptation of CPS under different physical activities, create the design framework of CPS for blind individuals, and deploy CPS systems for everyday use in different physical activities.

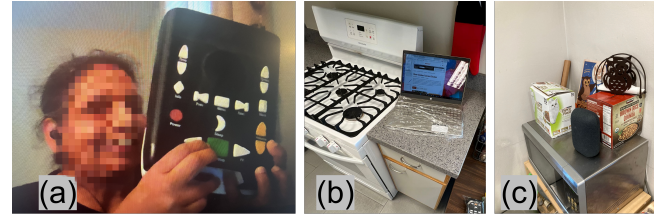


Figure 1: Practices of Technology Adaptation for Cooking.

3 METHODOLOGY

3.1 Understanding Adaptation and Design Framework

To explore the adaptation of CPS by people with vision impairments in different physical activities, we aim to first conduct user-centric mixed-method studies with blind individuals. The approaches include content analysis through online platforms [20], survey [22], and semi-structured interviews [20]. Based on the findings of existing practices and challenges of CPS (Figure 1), we further explore the design guidelines and framework of CPS for people with vision impairments through further semi-structured interviews and co-design workshops. We create design probes to guide and inspire our participants throughout the design process.

3.2 Design Integration and Feasibility

Based on the design guidelines and framework of CPS for specific physical activities, we develop CPS as a technology probe to further uncover the practices and challenges of the CPS by people with vision impairments through deployment. Specifically, the main difficulty of exploring and interacting with physical space (e.g., kitchen) for people with vision impairments is the lack of environmental awareness [20]. Body-worn cameras, as one of the representatives of CPS, have long been explored the benefits of recognizing objects surrounding the person without user intervention [16, 19] (Figure 2). We develop body-worn cameras to understand the feasibility of leveraging the implicit capturing to recognize objects in the physical space [16, 19].

3.3 High-Fidelity Implementation for Deployment [Planned]

Given different people different form factors of body-worn cameras, we plan to build an attachable Pi camera along with Raspberry Pi 4, a Lithium battery for object recognition, and a laptop. The attachable Pi camera will be designed to attach to different positions based on user experience, such as glasses, neckless, watch, and helmet. We aim to understand and explore how to leverage existing and non-perfect object recognition algorithms (e.g., YOLO [3]) to



Figure 2: (a) Example of body-worn camera [19]. (b) Image captured through body-worn cameras [16, 19].

provide useful information to people with vision impairments for environmental awareness (Figure 3). The system will also involve a voice-based LLM conversational agent (e.g., GPT-4 [2]) that allows the user to ask questions regarding the contextual information of the objects. The exploration involves usability considerations for body-worn cameras at different positions by blind individuals, information needs by blind people to acquire the object information (e.g., shape, color, spatial location), and feedback modes for CPS to blind people while enhancing environmental awareness.



Figure 3: Expected attachable body-worn cameras.

4 EVALUATION

After building the system, we will conduct user studies with blind individuals. The user studies will be conducted in-situ in a home environment. We will misplace certain objects that participants are familiar with at different locations. The user study will evaluate the following:

- **System Performance:** We are interested to explore the overall performance of the body-worn camera in different positions and why certain objects were not recognized. This

would bring insights to future researchers on how different positions of body-worn cameras may impact specific types of interactions or objects.

- **Object of Interest in Physical Activities:** With the deployment of our attachable body-worn camera, we are interested in further understanding the object of interest in physical activities through the in-situ deployment study. During the study, we will prompt questions to the participant regarding what types of objects they are interested in. This also includes shapes, colors, and spatial locations.
- **Feedback Mode:** Inspired by prior research of audio guide to blind individuals [14], participants will be asked to use two feedback modes (i.e., active and passive) to understand the verbosity of audio feedback of CPS to support people with vision impairments in physical space.
- **Social Perception:** According to Shinohara and Wobbrock [26], we are interested to understand the social perceptions of both the blind individuals with body-worn cameras and other people who observed the user study regarding how obtrusive the system is and potential social misperceptions that such system might generate. We will work with the participants to reduce unwanted attention and improve the overall social acceptance of the CPS in physical activities.
- **Usability Improvements for Deployment:** After the in-situ study, we will conduct a follow-up interview to explore the overall usability improvement of the CPS and further generate design guidelines and considerations for the deployment of CPS for people with vision impairments.

5 CONTRIBUTION

Our research first contributes to the understanding of existing practices, challenges, and strategies of adaptation of CPS by people with vision impairments. This could further guide future research to develop CPS to solve existing challenges by people with vision impairments (e.g., [20, 22]). Based on the existing adaptation of CPS, my research further introduces the design framework of CPS through

user-centered approach to improve the overall adaptation and social acceptance of CPS in physical activities. Then, my research further contributes to the integration of CPS and the deployment process. Overall, my research provides a roadmap of augmenting accessible CPS for people with disabilities in the physical space.

6 AUTHOR BIOGRAPHY

Franklin Mingzhe Li is an NSERC Postgraduate Fellow and a third-year PhD student in the Human-Computer Interaction Institute at Carnegie Mellon University. His PhD research is supervised by Dr. Patrick Carrington. Franklin started his PhD in September 2020 and expect to complete by September 2025. Franklin's research interests sit at the intersection of accessibility, HCI, and sensing. He is passionate about inclusive and universal design of technologies. He aims to develop and augment accessible CPS to support people with disabilities in physical space. He has published prior work in different journals (IWMUT, CSCW) and conferences (CHI, ASSETS, IUI, MobileHCI, DIS).

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